Perikymata Distribution from the Final Curve

Given the final outer enamel surface (OES) $\mathbf{r}_f(s) = (x_f(s), y_f(s))$ and the termination time field $t^*(s)$, the perikymata distribution measures how much developmental time is compressed into each spatial region of the OES.

1 Establish a Reference Axis

In general, the reference axis is specified as an arbitrary unit vector $\hat{\mathbf{d}} \in \mathbb{R}^2$ and an origin point $\mathbf{r}_0$ on the OES. The projected coordinate of each point along the crown is then

$$\rho(s) = \big(\mathbf{r}_f(s) - \mathbf{r}_0\big) \cdot \hat{\mathbf{d}}. \tag{1}$$

The integration bounds are $\rho \in [0, H]$ where $H = \max_s \rho(s)$. When the geometry admits well-defined anatomical landmarks, $\hat{\mathbf{d}}$ and $\mathbf{r}_0$ can be set naturally from the cusp and cervix; in the general case they are free parameters that define the measurement orientation.

2 Partition the OES into Intervals

Divide $[0, H]$ into N equal intervals of width $\Delta = H/N$. Define a smooth indicator for the k-th interval:

$$\chi_k(s) = \sigma\big(\beta\big(\rho(s) - k\Delta\big)\big) \cdot \sigma\big(\beta\big((k+1)\Delta - \rho(s)\big)\big)\,, \tag{2}$$

where σ is the logistic function and β controls sharpness.

3 Integrate Developmental Time per Interval

$$\mathcal{T}_k = \int_0^L \left|\frac{dt^*}{ds}\right| \chi_k(s)\, ds \;\approx\; \sum_i dt_i^* \cdot \chi_k(s_{i+1/2}), \tag{3}$$

where $dt_i^* = |t^*(s_{i+1}) - t^*(s_i)|$.

4 Convert to Perikymata Count

Given Retzius periodicity $\mathcal{R}$ (days):

$$P_k = \frac{365\,\mathcal{T}_k}{\mathcal{R}}. \tag{4}$$

Non-uniform $t^*(s)$ directly produces non-uniform P_k, since regions where the off-front decelerates accumulate more dt^* per unit projected length.